

Lab Report: Week 9 - Acoustic Resonance

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Imports and Functions

All fits are straight lines by orthogonal distance regression, so the uncertainty on both axes is included.

```
from dataclasses import dataclass
from math import pi, log
import matplotlib.pyplot as plt
import numpy as np
from scipy import odr
from uncertainties import ufloat
```

```
@dataclass
```

```
class odr_fit_result:
    parameters: np.ndarray
    parameter_errors: np.ndarray
    r_squared: float
    x_nom: np.ndarray
    y_nom: np.ndarray
    x_err: np.ndarray
    y_err: np.ndarray
```

```
def linear_model(p, x):
    return p[0] * x + p[1]
```

```
def proportional_model(p, x):
    return p[0] * x
```

```

def odr_fit(x, y, model_func, beta0):
    x_nom, x_err = np.array([v.n for v in x]), np.array([v.s for v in x])
    y_nom, y_err = np.array([v.n for v in y]), np.array([v.s for v in y])

    data = odr.RealData(x_nom, y_nom, sx=x_err, sy=y_err)
    odr_result = odr.ODR(data, odr.Model(model_func), beta0=beta0).run()

    residuals = y_nom - model_func(odr_result.beta, x_nom)
    r_squared = 1 - np.sum(residuals**2) / np.sum((y_nom - np.mean(y_nom)) ** 2)

    return odr_fit_result(odr_result.beta, odr_result.sd_beta, r_squared, x_nom,
                           y_nom, x_err, y_err)

def sigma_test(k1, sk1, k2, sk2, label="parameter"):
    sigma_diff = abs(k1 - k2) / np.sqrt(sk1**2 + sk2**2)
    print(f"\nSIGMA TEST ({label})")
    print(f"value 1 : {k1:.3f} +/- {sk1:.3f}")
    print(f"value 2 : {k2:.3f} +/- {sk2:.3f}")
    print(f"discrepancy: {sigma_diff:.2f} sigma -> {'AGREE' if sigma_diff <= 3
else 'DISAGREE'}")
    return sigma_diff

def plot_fit(result, model_func, xlabel, ylabel, title, filename):
    x_smooth = np.linspace(min(result.x_nom), max(result.x_nom), 200)
    residuals = result.y_nom - model_func(result.parameters, result.x_nom)

    fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(7, 7), sharex=True,
    gridspec_kw={"height_ratios": [3, 1]})
    ax1.errorbar(result.x_nom, result.y_nom, xerr=result.x_err,
    yerr=result.y_err, fmt="o", capsize=3, label="Data")
    ax1.plot(x_smooth, model_func(result.parameters, x_smooth), "k--",
    label=f"ODR fit ($R^2$={result.r_squared:.4f})")
    ax1.set_ylabel(ylabel)
    ax1.set_title(title)
    ax1.grid(True, alpha=0.6)
    ax1.legend()

    ax2.errorbar(result.x_nom, residuals, yerr=result.y_err, fmt="o", capsize=3,
    color="crimson")
    ax2.axhline(0, color="black", linestyle="--")
    ax2.set_xlabel(xlabel)

```

```
ax2.set_ylabel("Residual")
ax2.grid(True, alpha=0.6)

plt.tight_layout()
plt.savefig(f"graphs/{filename}.png", dpi=300)
plt.close()
```

Pre-Lab Questions

Question 2

At the closed end of the pipe, air molecules cannot move freely, creating a displacement node (pressure antinode). At the open end, the air pressure is constrained to equal atmospheric pressure, creating a pressure node (displacement antinode).

The effective length of a pipe closed at one end is

$$L_{\text{eff}} = L + 0.6a$$

standing waves inside a pipe closed at one end = odd number of $\frac{\lambda}{4}$

$$L + 0.6a = \frac{n\lambda}{4}$$

$$\lambda_n = \frac{4(L + 0.6a)}{n}$$

using $f = \frac{c}{\lambda}$,

$$f_n = \frac{nc}{4(L + 0.6a)}$$

$$f_n = \frac{nc}{4(L + 0.6a)}, \quad n = 1, 3, 5, \dots$$

Question 3

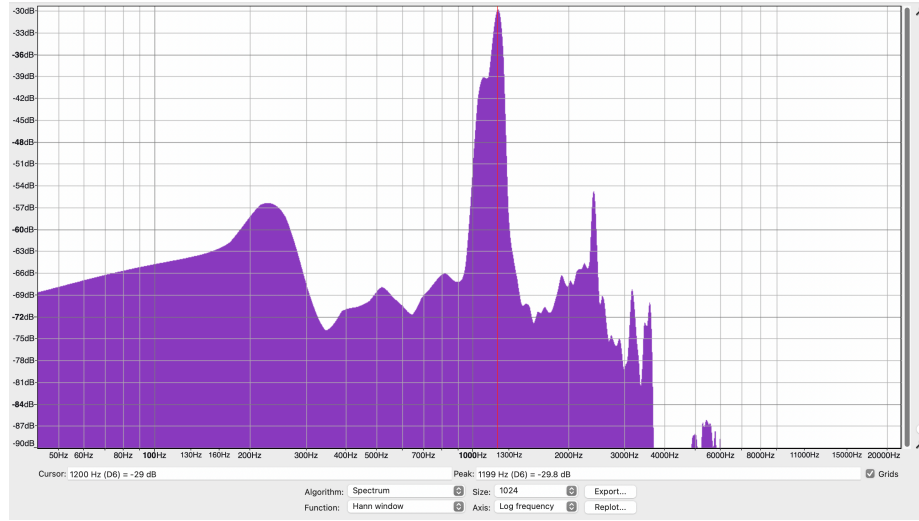


Figure 1: spectrum plotted

reading off the graph and the peak indicator, we see that the highest peak is at 1199 Hz.

$$f_{\text{peak}} = 1199 \text{ Hz}$$

Aim

To find the speed of sound from the resonances of a pipe open at both ends and of a pipe closed at one end, and to measure the half-life and quality factor of the first three resonances of the open pipe.

Theory

A pipe open at both ends has a pressure node about $0.6a$ beyond each end (a is the pipe radius), so L becomes $L + 1.2a$. The closed-open pipe has only one open end and is derived in the pre-lab:

$$f_n = \frac{nc}{2(L + 1.2a)} \quad (\text{open-open}), \quad f_n = \frac{(2n-1)c}{4(L + 0.6a)} \quad (\text{closed-open})$$

Both have the form $f = cx$, so a plot of frequency against x is a line through the origin with slope c . The theoretical speed of sound in dry air at T ($^{\circ}\text{C}$) is $c = 331.3 + 0.606 T$ m/s.

Resonance intensity decays as $I_0 e^{-\alpha t}$, a drop of 3 dB per half-life $T_{1/2}$, so the level falls linearly, $L(t) = L_0 - 3t/T_{1/2}$ and $T_{1/2} = 3/(-m)$ for slope m . The quality factor is $Q = \pi f_r T_{1/2} / \ln 2$.

Two values are compared as $|k_1 - k_2| / \sqrt{\sigma_1^2 + \sigma_2^2}$, and agree if this is at most 3σ .

Procedure

First, we set up the PVC pipe like in the diagram, and taped the joint in the middle twice so air wouldn't leak. Then we measured the pipe's length, $L = 100.4 \pm 0.25$ cm, and radius, $a = 4.3 \pm 0.25$ cm, and estimated the room temperature, $T = 22 \pm 1$ °C.

For the first part, we tested the open end. We clamped the pipe horizontally to a table and positioned a microphone right near the end. We recorded acoustic responses generated by claps, physical taps, and loud white noise played from a phone at the far end. We repeated the exact same procedure using the closed-end cap (which had a small pre-drilled hole for the mic). In Audacity, each recording was analysed with Plot Spectrum and the peak frequencies were read with the cursor. Each peak was taken to be uncertain by $\sigma_f = 15$ Hz, about a third of the width of the bins.

for the decay, we used the recording from the first open-end clap experiment, with the first window starting at the clap and 30 ms windows after it. The level of each mode in a window was taken as the band level within ± 25 Hz of the mode.

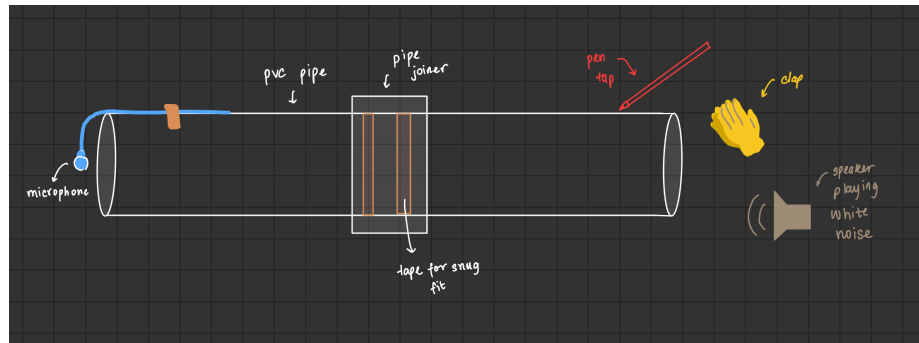


Figure 2: Experimental setup

Results

Choice of excitation method

The number of modes that could be read from each method is below. The clap gave the most modes on both pipes, so it was used as the best method and for the decay. The white-noise spectrum of the open pipe had peaks below 700 Hz that probably came from the phone speaker and not the pipe, so only modes 5 to 7 were used from it.

Method	Open pipe modes	Closed pipe modes
Clap	6 (1 to 6)	7 (2 to 8)
Tap	3 (1 to 3)	5
White noise	3 (5 to 7)	7 (2 to 8)

Speed of sound

Each recording (clap, tap, white noise on each pipe) is fitted for its own c , and the six are averaged and compared with the theoretical value.

```
L = ufloat(1.004, 0.0025)    # m, pipe length
a = ufloat(0.043, 0.0025)    # m, pipe radius
sigma_f = 15                # Hz, uncertainty on one peak frequency
T_room = ufloat(22, 1)       # deg C

open_data = [
    {1: 166.6, 2: 326.5, 3: 465.6, 4: 655.1, 5: 817.8, 6: 982.3},
    {1: 161.0, 2: 322.1, 3: 472.3},
    {5: 819, 6: 982, 7: 1125},
]

closed_data = [
    {2: 258.3, 3: 413.9, 4: 595.5, 5: 758.9, 6: 931.2, 7: 1098.0, 8: 1263.7},
    {2: 257.4, 3: 415.7, 4: 578.3, 6: 930.0, 8: 1240.7},
    {2: 259, 3: 419, 4: 590, 5: 753, 6: 930, 7: 1110, 8: 1252},
]

# f = c * x
x_open = lambda modes: [n / (2 * (L + 1.2 * a)) for n in modes]
x_closed = lambda modes: [(2 * n - 1) / (4 * (L + 0.6 * a)) for n in modes]

c_methods = []
for pipe, data, x_func in [("open", open_data, x_open), ("closed", closed_data,
x_closed)]:
    for label, d in zip(["clap", "tap", "white noise"], data):
        modes = sorted(d)
        f = [ufloat(d[n], sigma_f) for n in modes]
        res = odr_fit(x_func(modes), f, proportional_model, beta0=[340])
        c = ufloat(res.parameters[0], res.parameter_errors[0])
        print(f"{pipe} {label}: c = {c:.2f} m/s, R^2 = {res.r_squared:.4f}")
        if label == "clap":    # the three methods give near-identical graphs,
so only the clap is plotted
            plot_fit(res, proportional_model, "x (1/m)", "Resonant frequency (Hz)",
                    f"{pipe.capitalize()} pipe, clap", f"speed_{pipe}_clap")
            c_methods.append(c)

c_open, c_closed, c_mean = sum(c_methods[:3]) / 3, sum(c_methods[3:]) / 3,
sum(c_methods) / 6
print(f"\nopen pipe average: c = {c_open:.2f} m/s")
print(f"closed pipe average: c = {c_closed:.2f} m/s")
```

```

print(f"average of all six: c = {c_mean:.2f} m/s")

c_theory = 331.3 + 0.606 * T_room    # m/s
print(f"theoretical c at {T_room} C = {c_theory:.2f} m/s")
sigma_test(c_open.n, c_open.s, c_closed.n, c_closed.s, label="open vs closed")
sigma_test(c_mean.n, c_mean.s, c_theory.n, c_theory.s, label="average c vs theory")

```

```

open clap: c = 343.77+/-2.44 m/s, R^2 = 0.9987
open tap: c = 335.11+/-2.58 m/s, R^2 = 0.9991
open white noise: c = 342.86+/-2.23 m/s, R^2 = 0.9948
closed clap: c = 347.71+/-0.74 m/s, R^2 = 0.9998
closed tap: c = 343.20+/-1.86 m/s, R^2 = 0.9995
closed white noise: c = 347.14+/-1.35 m/s, R^2 = 0.9994

```

```

open pipe average: c = 340.58+/-1.40 m/s
closed pipe average: c = 346.02+/-0.80 m/s
average of all six: c = 343.30+/-0.81 m/s
theoretical c at 22.0+/-1.0 C = 344.63+/-0.61 m/s

```

```

SIGMA TEST (open vs closed)
value 1 : 340.582 +/- 1.398
value 2 : 346.016 +/- 0.803
discrepancy: 3.37 sigma -> DISAGREE

```

```

SIGMA TEST (average c vs theory)
value 1 : 343.299 +/- 0.806
value 2 : 344.632 +/- 0.606
discrepancy: 1.32 sigma -> AGREE

```

```

np.float64(1.3217733240226746)

```

Pipe	Method	c (m/s)
Open	clap	343.77 ± 2.44
Open	tap	335.11 ± 2.58
Open	white noise	342.86 ± 2.23
Closed	clap	347.71 ± 0.74
Closed	tap	343.20 ± 1.86
Closed	white noise	347.14 ± 1.35

Pipe	Method	c (m/s)
Open average		340.58 ± 1.40
Closed average		346.02 ± 0.80
Average of all six		343.30 ± 0.81

The average of all six agrees with the theoretical 344.63 ± 0.61 m/s at 22 ± 1 °C (1.32σ). The open and closed averages differ by 3.37σ , just past the 3σ cut-off, so the two pipes do not quite agree within error.

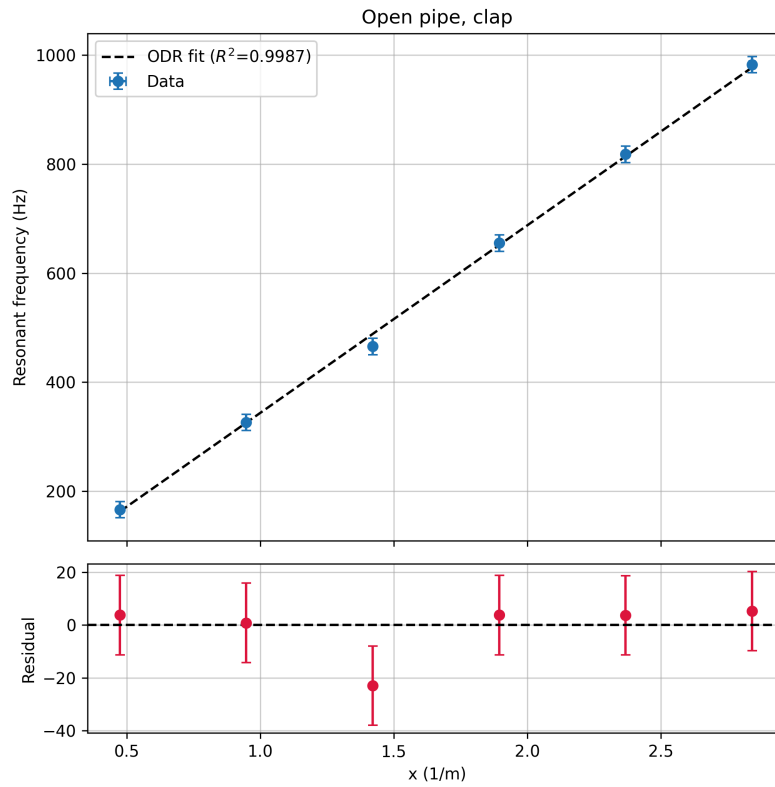


Figure 3: Open pipe, clap: fit and residuals

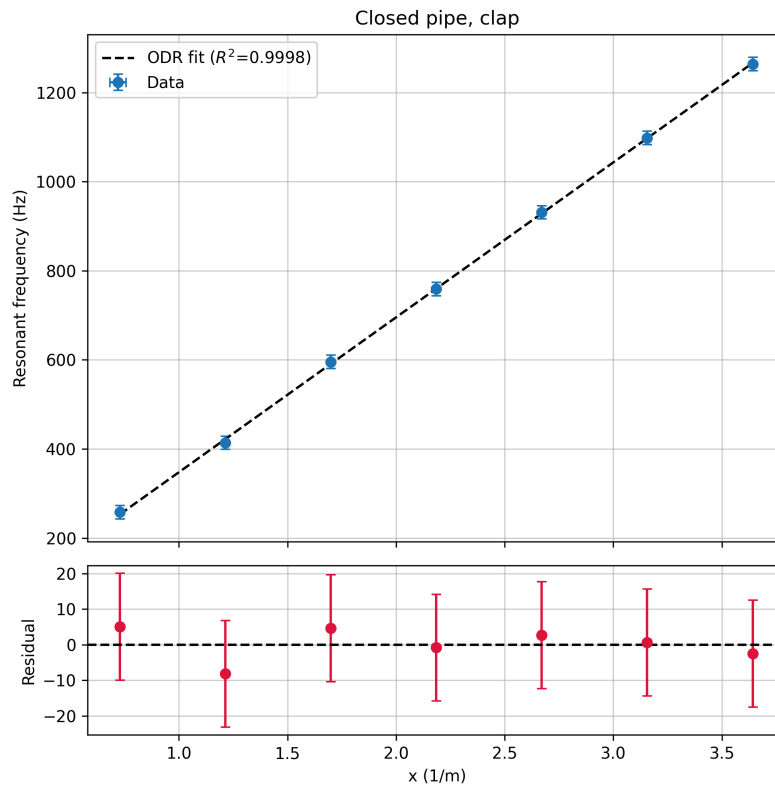


Figure 4: Closed pipe, clap: fit and residuals

Resonance decay

```

decay_freqs = [166.6, 326.5, 465.6] # Hz, open clap modes 1-3
Dt = 0.03 # s, window length
sigma_t = 0.002 # s, uncertainty on each t_n
sigma_L = 2.0 # dB, uncertainty on one level
# level (dB) of each mode in window n, t_n = n * Dt (band level within +/-25 Hz
of the mode)
decay_L = [
    [-26.8, -28.1, -31.3, -43.7, -46.0, -53.8, -56.2],
    [-23.5, -25.2, -35.7, -37.8, -44.4, -47.0],
    [-27.8, -29.5, -36.2, -32.4, -41.7],
]

for i in range(3):
    t = [ufloat(n * Dt, sigma_t) for n in range(len(decay_L[i]))]
    L_db = [ufloat(v, sigma_L) for v in decay_L[i]]
    res = odr_fit(t, L_db, linear_model, beta0=[-100, 0])
    slope = ufloat(res.parameters[0], res.parameter_errors[0])
    T_half = 3 / (-slope)

```

```

    Q = pi * decay_freqs[i] * T_half / log(2)
    print(f"\nmode {i + 1} ({decay_freqs[i]} Hz): slope = {slope:.1f} dB/s, R^2
= {res.r_squared:.4f}, "
        f"T_1/2 = {T_half * 1000:.1f} ms, Q = {Q:.1f}")
    plot_fit(res, linear_model, "$t_n$ (s)", "Level (dB)", f"Decay of mode {i +
1}", f"decay_mode{i + 1}")

```

```

mode 1 (166.6 Hz): slope = -184.0+/-17.6 dB/s, R^2 = 0.9563, T_1/2 = 16.3+/-1.6
ms, Q = 12.3+/-1.2

```

```

mode 2 (326.5 Hz): slope = -169.0+/-17.0 dB/s, R^2 = 0.9609, T_1/2 = 17.8+/-1.8
ms, Q = 26.3+/-2.6

```

```

mode 3 (465.6 Hz): slope = -102.7+/-33.3 dB/s, R^2 = 0.7587, T_1/2 = 29.2+/-9.5
ms, Q = 61.7+/-20.0

```

Mode	f_r (Hz)	$T_{1/2}$ (ms)	Q	R^2
1	166.6	16.3 ± 1.6	12.3 ± 1.2	0.956
2	326.5	17.8 ± 1.8	26.3 ± 2.6	0.961
3	465.6	29 ± 10	62 ± 20	0.759

Modes 1 and 2 have about the same half-life, so Q roughly doubles with frequency. Mode 3 has a longer half-life but a large uncertainty and a poor fit, and is within 1.2σ of mode 2.

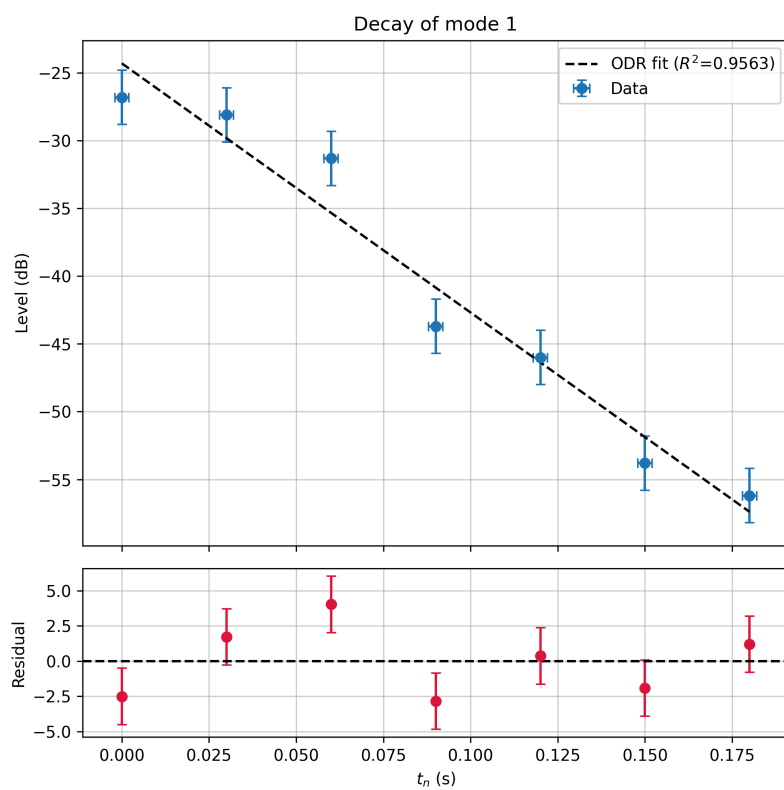


Figure 5: Decay of mode 1: fit and residuals

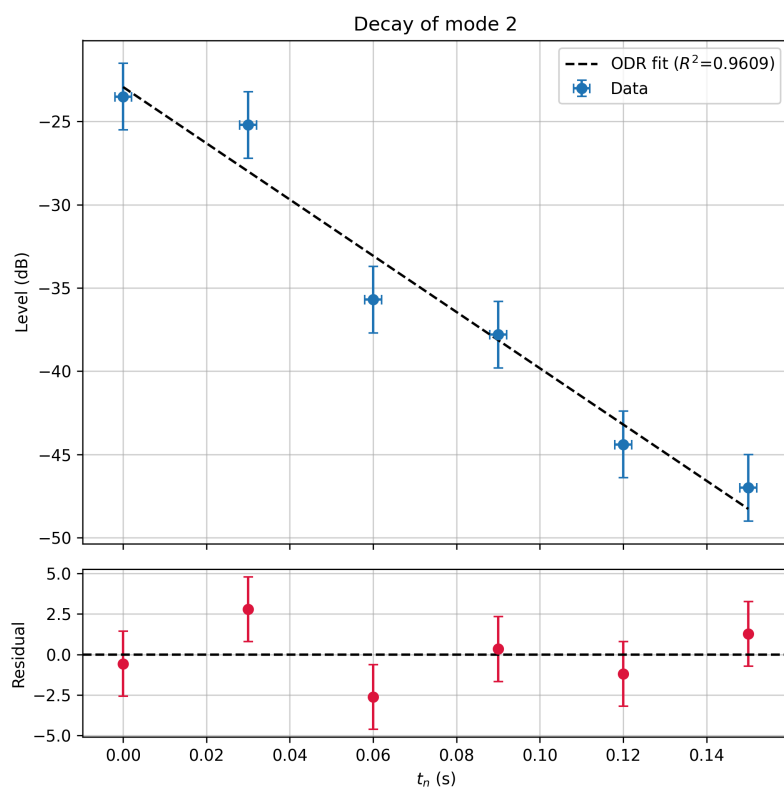


Figure 6: Decay of mode 2: fit and residuals

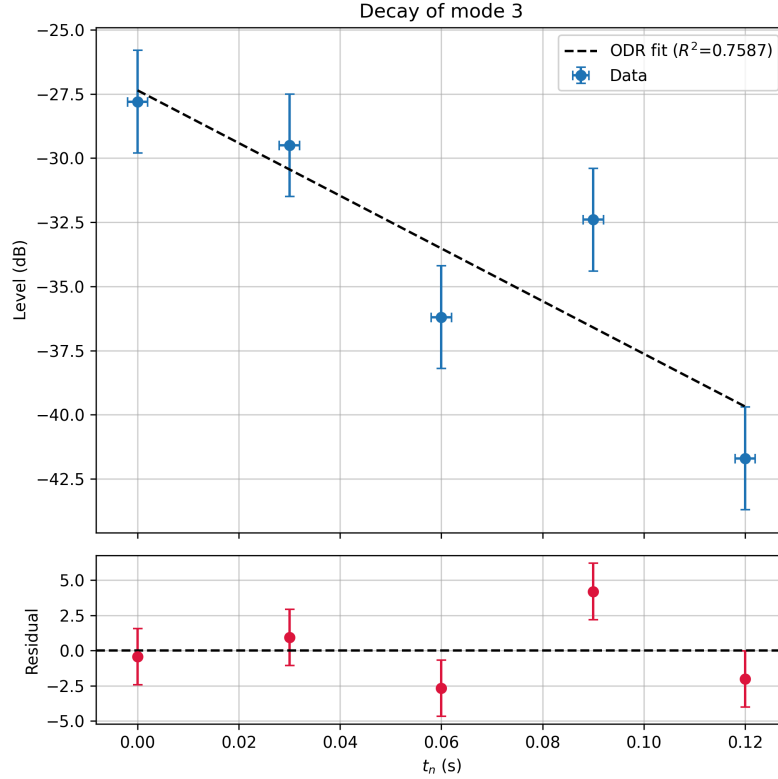


Figure 7: Decay of mode 3: fit and residuals

Limitations & Possible Sources of Discrepancy

Several steps were taken to reduce the uncertainties. The joint was taped twice to avoid leaks and reflections, and the end corrections were included when calculating the effective length. Each pipe was measured with three excitation methods, and the speed of sound was taken from fits through many modes (up to 8) rather than from one, using orthogonal distance regression so the uncertainty in L and a is included, and the six results were averaged, giving a smaller uncertainty (± 0.81 m/s) than most of the individual ones. Readings that were not pipe modes, namely the white-noise speaker peaks and the 90 to 110 Hz hump on the closed pipe, were excluded so that they did not bias c . For the decay, the windows started at the clap, only levels above the background were fitted. Further improvement would come from reading the frequencies at a higher resolution to shrink σ_f , measuring L and a with finer instruments and repeating each recording.

1. Peak frequencies: $\sigma_f = 15$ Hz is an estimate, and the spectra were read at coarse resolution. The open tap (335.11 ± 2.58 m/s) and closed clap (347.71 ± 0.74 m/s) are each more than 3σ from theory, on opposite sides, so the uncertainties on c are probably underestimated.

2. Shared errors: The six methods share L , a and the pipe, so the uncertainty on their average is optimistic. A slight error in L alone changes our values completely.

3. Closed pipe mode 1: The expected fundamental of about 84 Hz was left out, because the 90 to 110 Hz hump appears in every spectrum, including the open pipe's.

4. Decay: With 30 ms windows each fit has only 5 to 7 points, and the mode 3 fit is poor.

Conclusion

The speed of sound was measured from the resonances of a pipe open at both ends and of a pipe closed at one end, using a clap, a tap and white noise on each. The average over the six measurements is 343.30 ± 0.81 m/s, which agrees with the theoretical 344.63 ± 0.61 m/s at 22° C (1.32σ). The open pipe average (340.58 ± 1.40 m/s) and closed pipe average (346.02 ± 0.80 m/s) differ by 3.37σ , so the two pipes only just fail to agree within error. The most likely cause is that the peak frequency uncertainty is underestimated, since the open tap and closed clap results each sit more than 3σ from theory on opposite sides.

The first three resonances of the open pipe decay with half-lives of 16.3 ± 1.6 , 17.8 ± 1.8 and 29 ± 10 ms, giving quality factors of 12.3 ± 1.2 , 26.3 ± 2.6 and 62 ± 20 . Q rises with frequency because the first two half-lives are similar and $Q \propto f_r T_{1/2}$. The mode 3 value is the least reliable ($R^2 = 0.76$).